

Lecture 22: Clinical Imaging — Modeling

BINF 4002 — Machine Learning for Health

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Modality progression:

- L17–18: EHR & Claims • L19–20: Clinical Text & NLP
- L21 (Tue): Clinical Imaging — **Data**
- L22: Clinical Imaging — **Modeling** ← **today**
- L23+: Population Health, Omics, Multimodal

Tuesday: what medical images *are*—physics, economics, metadata, labels.

Today: how to *model* them—and where standard ML approaches break.

Reorganized around what's unique here:

You know CNNs, encoders, attention. We won't re-teach those. Instead: why does label quality dominate? Why does segmentation need a new architecture? Why does pathology need a new problem formulation? Why do standard augmentations break?

Recall: The Five-Question Framework (L21)

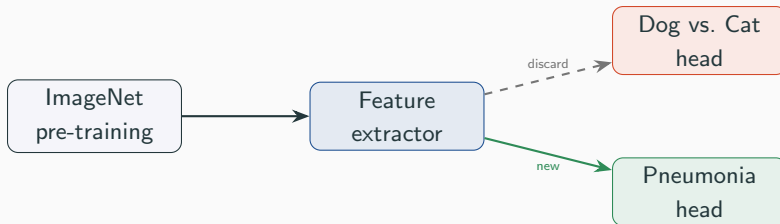
Tuesday's framework—today we address the **modeling consequences**:

1. **Physics** → determines whether transfer learning from natural images makes sense
2. **Acquisition** → determines dataset size, which drives pre-training strategy
3. **Metadata** → scanner variation causes distribution shift across sites
4. **Labels** → NLP-extracted silver-standard labels set a noise ceiling
5. **Pitfalls** → shortcuts, augmentation validity, evaluation design

Today's theme: In most medical imaging settings, data quality, labeling, and evaluation are the *first* bottlenecks. Architecture still matters—but usually after those are under control.

Transfer Learning

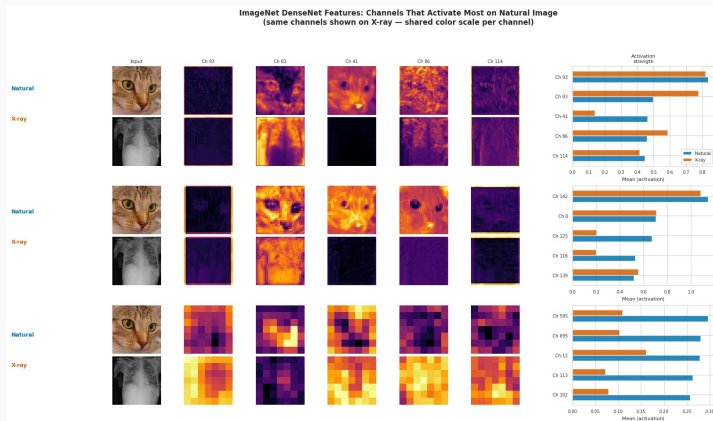
Transfer Learning: The Standard Recipe



Standard assumption: low-level features transfer across domains. **But does this actually help?**

Notebook Fig. 1 compares DenseNet features on a natural image vs. a chest X-ray.

ImageNet Features: Natural vs. Medical (Notebook Fig. 1)



We pick channels that activate *most strongly on the cat* and show those same channels on the X-ray. **Early:** both images activate similarly (edges are universal). **Late:** the cat's top channels are weak on the X-ray—the network doesn't find what it was trained to detect.

Does ImageNet Transfer Actually Help?

Raghu et al., 2019 (“Transfusion”):

- Evaluated on retinal disease & chest X-ray

Why? Recall L21 (physics):

- Natural: global subjects, RGB, occlusion
- Medical: **local texture**, grayscale, standardized
- Only lowest layers retain ImageNet features

Takeaway

Don't assume transfer helps.

Measure it. Consider smaller models.

Previous slides: late features activate weakly on X-rays. Notebook Fig. 2: training curves.

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- Higher layers barely differ from random init

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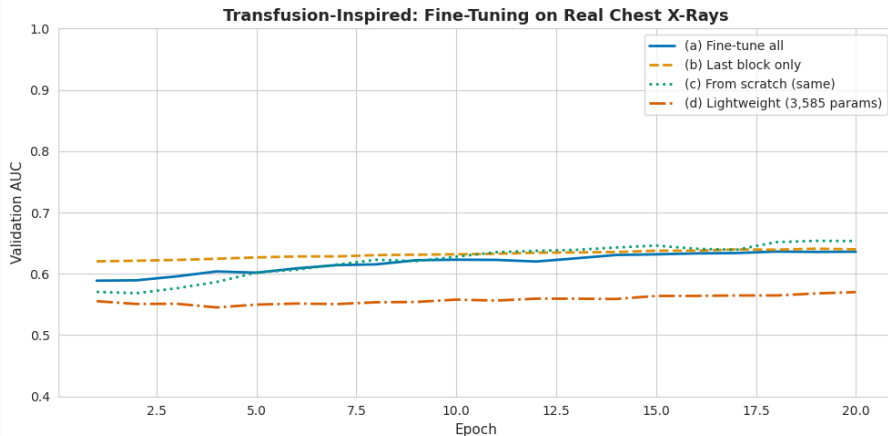
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Transfusion Results on Real X-Rays (Notebook Fig. 2)



Strategies (a)–(c) converge to similar final AUC despite different initialization. The lightweight model (d) is lower—model capacity matters, but pre-training doesn't. This is the Transfusion finding on real ChestMNIST data.

When Does Pre-training Help?

Domain-specific pre-training:

- Train on large medical corpora *before* fine-tuning
- Self-supervised: SimCLR, DINO, MAE—no labels needed
- Notable: CheXzero, REMEDIS, CONCH, BiomedCLIP

Helps Most When...

- Labeled data is small (< 1000)
- Domain features matter (HU, stain)
- Task is “far” from ImageNet

Matters Less When...

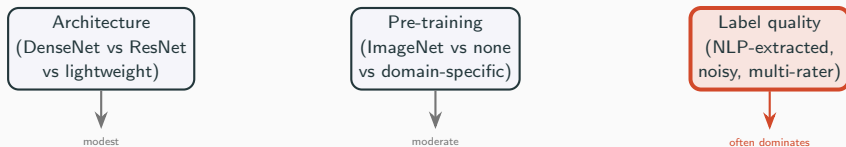
- Large labeled datasets available
- Models already over-parametrized

Recall L21 (acquisition): MRI/pathology are expensive $\rightarrow$ small data $\rightarrow$ pre-training helps. X-ray is cheap $\rightarrow$ large data $\rightarrow$ less benefit.

The Label Quality Problem

Label Quality: The Binding Constraint

This is **not** a classification-specific problem—it affects every imaging task.



Recall L21 (labels): Image $\rightarrow$ radiologist $\rightarrow$ report $\rightarrow$ NLP $\rightarrow$ label. Each step adds noise. Expert disagreement alone can be substantial. You are training on *labels that are themselves model outputs*.

Practical implication: Invest in label quality (adjudication, consensus reads, cleaning NLP labels) rather than architecture search. This applies to classification, segmentation, and MIL equally.

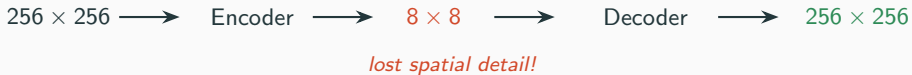
Segmentation & U-Net

Segmentation: The Clinical Need

Goal: Assign a class label to every pixel (or voxel).

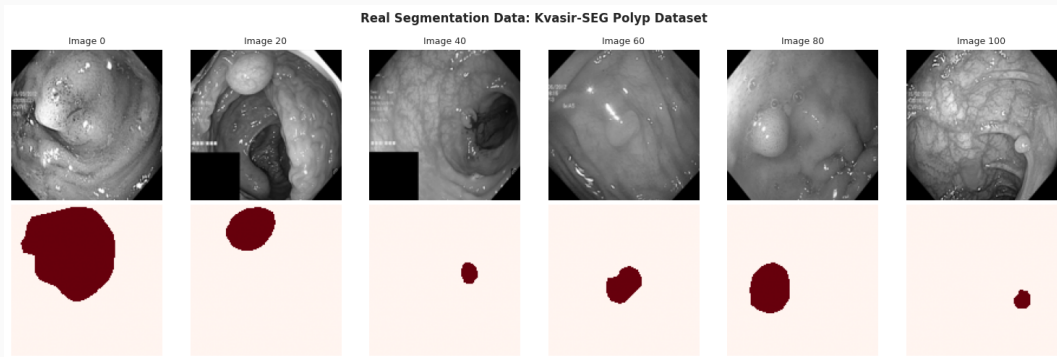
Clinical applications: Tumor delineation for radiation therapy, organ segmentation for surgical planning, lesion measurement for response tracking.

Key challenge: A standard encoder compresses spatial information—but segmentation needs to recover the original resolution.



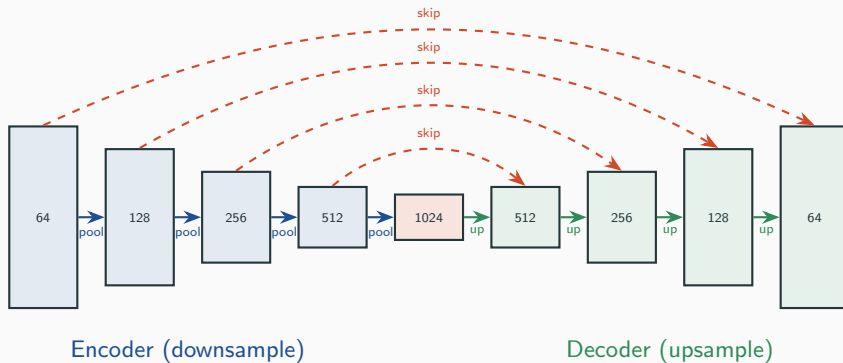
Why not keep full resolution? A 512×512 feature map with 256 channels = 256 MB per layer. With 10 layers, you can't fit even one image in GPU memory. Downsampling is necessary—the question is how to recover what you lose.

Real Segmentation Data: Kvasir-SEG



1,000 colonoscopy images with expert polyp masks (Jha et al., 2020). Notice the range of polyp sizes—from large (left) to tiny (right). Small polyps occupy $<1\%$ of pixels, making CE loss ineffective. This is the data we train on in the notebook.

U-Net: Born in Biomedical Imaging



Skip connections **concatenate** encoder features with decoder features at each resolution.

Ronneberger et al., 2015 — designed for biomedical segmentation with limited data. Dominant 10+ years later. Notebook Fig. 3: U-Net vs. no-skip baseline.

Skip Connections: U-Net vs. ResNet

You've seen skip connections before—**residual connections** in ResNet. How do they differ?

ResNet (residual):

- $x_{\text{out}} = F(x) + x$ (additive)
- Same resolution throughout
- Purpose: gradient flow in deep networks
- Skip carries the *same* features

U-Net (skip):

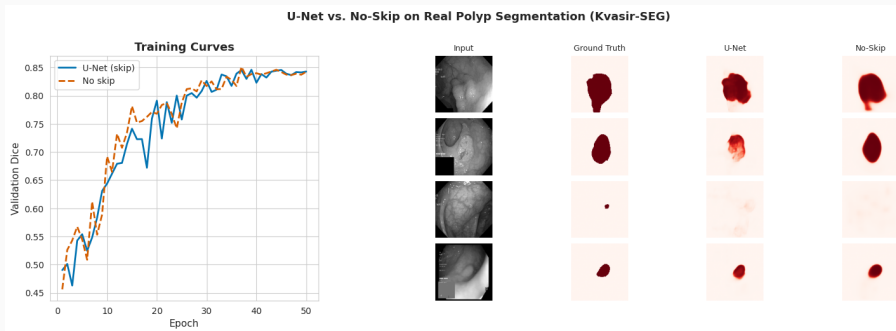
- $x_{\text{out}} = G([x_{\text{up}} \parallel x_{\text{enc}}])$ (concat)
- *Across* resolutions (enc $\rightarrow$ dec)
- Purpose: recover spatial detail lost by pooling
- Skip carries *different* features (high-res)

Why this matters clinically:

- Without skip: blurry boundaries, small structures lost
- With skip: sharp boundaries—radiation planning requires sub-mm precision

Variants: 3D U-Net (volumetric), Attention U-Net, nnU-Net (“AutoML for segmentation”).

Skip Connection Comparison (Notebook Fig. 3)



Real Kvasir-SEG polyps. Dice may be similar—polyps are large. Examine *boundaries*: skip connections preserve edge detail that Dice is insensitive to.

Dice Loss: What $\hat{y}$ Means

Problem: Tumor occupies $< 1\%$ of pixels. CE loss: 99% of the gradient says “predict background.”

Dice loss: Normalizes by *region size*, not pixel count.

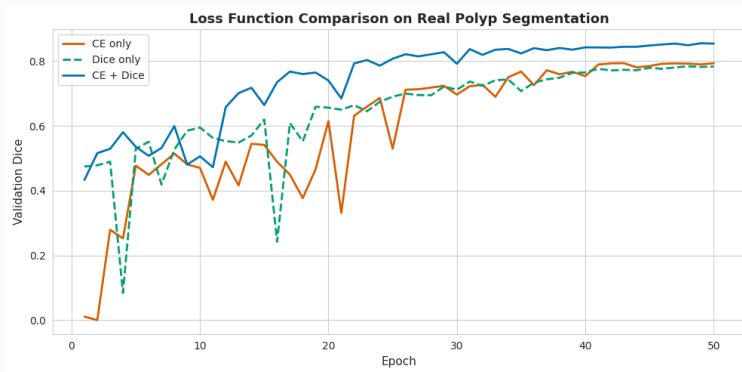
$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i y_i \hat{y}_i + \epsilon}{\sum_i y_i + \sum_i \hat{y}_i + \epsilon}$$

Important: $\hat{y}_i = \sigma(z_i) \in [0, 1]$ is a **probability** (sigmoid output), not a hard label. This makes the loss differentiable for training. At evaluation, you threshold ($\hat{y}_i > 0.5$) to get the Dice *coefficient* (= F1 on binary masks).

Why Dice focuses on the foreground: If the foreground has 100 pixels and you miss them all, $\sum y_i \hat{y}_i \approx 0$ so $\mathcal{L}_{\text{Dice}} \approx 1$ regardless of how many of the 9,900 background pixels you got right. CE loss would still be ≈ 0.01 —background dominates.

In practice: $\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}$. Notebook Fig. 4: experimental comparison.

Loss Function Comparison (Notebook Fig. 4)



CE+Dice (blue) converges highest and most stably. CE-only (orange) is noisy—gradient dominated by background. Dice-only (green) starts unstable but recovers. Combined loss gets the best of both: stable gradients from CE, foreground focus from Dice.

Whole-Slide Pathology & MIL

The Gigapixel Problem

Recall L21 (histopathology): WSIs are $\sim 100K \times 100K$ pixels.

- Cannot fit in GPU memory (a single slide is ~ 5 GB uncompressed)
- Labels exist only at the *slide level*: “this patient has cancer”
- You tile the slide into 256×256 patches and process each independently



MIL: Most Tile Labels Are Wrong

The key insight that makes pathology MIL different from standard classification:

A slide labeled “cancer-positive” might contain 10,000 tiles, of which *maybe 50* actually show cancer. The other 9,950 tiles show normal tissue, stroma, fat, or artifacts.

If you naively assign the slide label to every tile:

- 99.5% of your “positive” training labels are **factually wrong**

MIL formalization: A bag is positive if *at least one* instance is positive.

- Positive bag: $Y_{\text{bag}} = 1 \iff \exists i : y_i = 1$ (at least one cancerous tile)
- Negative bag: $Y_{\text{bag}} = 0 \iff \forall i : y_i = 0$ (all tiles are normal)

This is why you need a *pooling* strategy, not per-tile classification.

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- Standard classification on these tiles will learn to predict “cancer” from normal stroma

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MIL Aggregation Strategies

Three strategies:

- **Max-pool:** $h_{\text{bag}} = \max_i h_i$ (most “positive” tile wins)

Attention-MIL (Ilse et al., 2018):

1. Tile features: $h_i = f_{\theta}(x_i)$
2. Attention: $a_i = w^{\top} \tanh(Vh_i)$
3. Normalize: $\alpha_i = \text{softmax}(a_i)$
4. Aggregate: $h_{\text{bag}} = \sum_i \alpha_i h_i$
5. Classify: $\hat{Y} = \sigma(Wh_{\text{bag}} + b)$

Notebook Fig. 5: MIL simulation comparing pooling strategies.

Clinical value of attention:

- Overlay α_i as heatmap on slide
- High-attention $\approx$ diagnostic regions
- **Caution:** artifacts (pen marks, folds) can dominate attention

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Attention-MIL (Ilse et al., 2018):

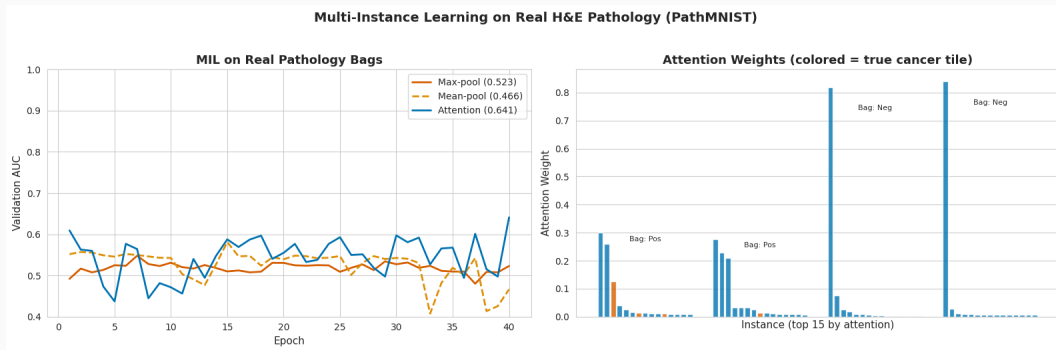
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MIL Aggregation Comparison (Notebook Fig. 5)



Left: bag-level AUC on real pathology bags. Attention-MIL learns to focus on the rare diagnostic tiles, outperforming both pooling baselines. Right: attention weights show concentration on specific tiles (tall bars) rather than uniform weighting—the model is learning *where* to look.

Augmentation: Why Physics Matters

Augmentation: Natural vs. Medical Images

The same augmentation can be valid for natural images and **invalid for medical images**—because the physics of acquisition differs.

Vertical flip:

- Natural image of a cat: still a cat → ✓
- Chest X-ray: heart now on the right → mimics dextrocardia (a rare clinical finding) → ✗
- Pathology tile: no canonical orientation → ✓

Principle: An augmentation is valid if and only if it produces images that *could plausibly occur* under the acquisition physics of that modality, without changing the clinical label.

Notebook Fig. 7 applies identical augmentations to a natural image, a chest X-ray, and a pathology tile side-by-side.

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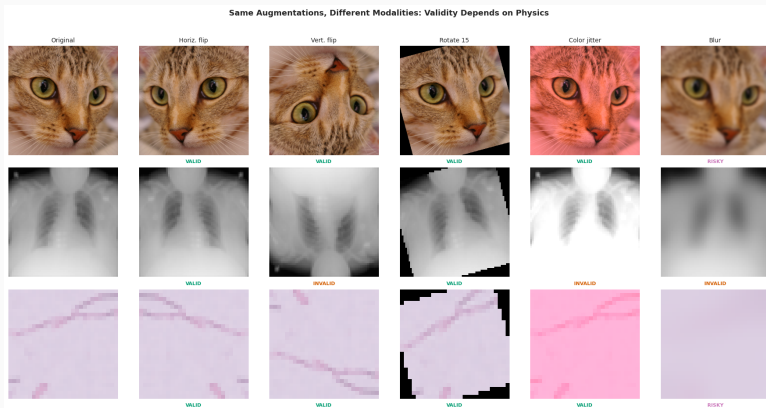
Color jitter:

- Natural image: simulates lighting variation → ✓
- X-ray: grayscale, intensity *is* the signal (tissue density) → ✗
- Pathology: simulates stain variation across labs → ✓

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Augmentation Validity: Real Images (Notebook Fig. 7)



Real images: natural (cat), chest X-ray (ChestMNIST), H&E pathology (PathMNIST). Vertical flip: valid for pathology, invalid for X-ray. Color jitter: valid for pathology (stain variation), invalid for X-ray (intensity is the signal).

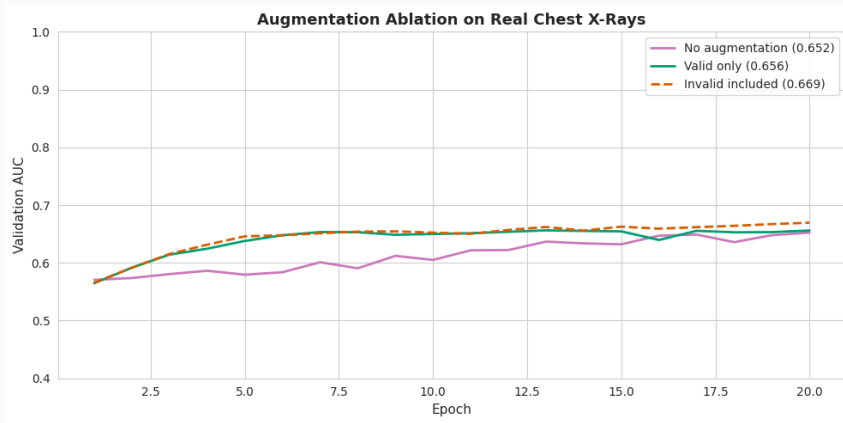
Augmentation: Modality-Specific Reference

Augmentation	Chest X-ray	Pathology	Retinal Fundus
Horizontal flip	✓ (usually; laterality can matter)	✓	✗ (OD/OS)
Vertical flip	✗ (up/down matters)	✓	✗
Rotation $\pm 15^\circ$	✓	✓	✓
Color jitter	~ (mild OK; intensity is signal)	✓ (stain var.)	✓ (camera)
Random crop	~ (may lose finding)	✓ (tiles)	~
Gaussian blur	✗ (hides fine detail)	~	✗

Lesson

Blindly applying a natural-image augmentation pipeline can *hurt* performance. Validate augmentations against clinical meaning. (Notebook Fig. 6)

Augmentation Ablation on Real X-Rays (Notebook Fig. 6)



No augmentation vs. valid-only (horiz. flip + rotation) vs. invalid included (adds vert. flip + intensity jitter). Even “invalid” augmentations can help via data volume—but risk encoding false invariances.

Deployment Pitfalls

Multi-Site Generalization

Recall L21 (metadata): Scanner manufacturer, field strength, and protocol differ across sites.

Site A
Scanner X
 $n = 10,000$

Site B
Scanner Y
 $n = 5,000$

Site C
Scanner Z
 $n = 2,000$

Model trained on A fails on C—not because the disease differs,
but because the **scanner** differs

Mitigation: Multi-site training, domain adaptation, external evaluation, per-site reporting.

Researcher degrees of freedom in medical imaging studies:

1. Dataset split (random vs. patient-level vs. temporal)

Each inflates performance by 1–5 AUC points. Compounded → “superhuman AI” hype.

Recall L21 (pitfalls): Shortcut learning (“PORTABLE” text) is a *predictable consequence* of the acquisition pipeline. The five-question framework helps anticipate these failures.

The Garden of Forking Paths

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Foundation Models

Foundation Models for Medical Imaging

The recent shift: Large-scale pre-trained models are changing the field.

Model	Modality	Method	Key Idea
CheXzero	Chest X-ray	CLIP (img+text)	X-rays paired with reports
CONCH	Pathology	Contrastive	Tissue images + captions
BiomedCLIP	Multi-modal	CLIP	Broad biomedical img+text
SAM-Med	Multi-modal	Promptable seg.	Segment anything in med imgs

What this changes for the five-question framework:

- **Transfer:** domain-specific pre-training at scale—addresses L21's physics gap

FMs don't solve the label quality or evaluation problems—they change *which* problems dominate.

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- **Transfer:** domain-specific pre-training at scale—addresses L21's physics gap
- **Labels:** zero-shot classification reduces labeling burden—but doesn't eliminate it
- **Pitfalls:** new failure modes (training data bias baked into the FM, harder to audit)

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Summary

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Paradigm	Task	Architecture	Key Challenge
Classification	What does image show?	CNN + fine-tuning	Label noise, evaluation
Segmentation	Where exactly?	U-Net + variants	Boundary precision, 3D
MIL	Slide-level from tiles	Attention-MIL	Tile labels are <i>wrong</i>

Decision rules for your next imaging project:

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- **Augmentation:** reason from physics, not from ImageNet defaults
- **Transfer:** measure the benefit; don't assume it helps

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- **Loss Functions:** CE from earlier; Dice loss new for segmentation

`nb22_medical_imaging_modeling.ipynb`:

1. ImageNet features on natural image vs. chest X-ray (Fig. 1)
2. Compare fine-tuning vs. lightweight models, Transfusion-inspired (Fig. 2)
3. Train U-Net vs. no-skip baseline on real polyp data (Kvasir-SEG, Fig. 3)
4. Compare loss functions: CE vs. Dice vs. combined (Fig. 4)
5. MIL on real H&E pathology tiles (PathMNIST, Fig. 5)
6. Augmentation ablation: valid vs. invalid transforms (Fig. 6)
7. Augmentation comparison: real medical vs. natural images (Fig. 7)

How to use: Run all cells, then experiment. All outputs save to `figures/`.

Questions?